Early Silurian paleolatitude for central Newfoundland from paleomagnetism of the Wigwam Formation[1]

KENNETH L. BUCHAN
Geological Survey of Canada, 601 Booth Street, Ottawa, Ont., Canada K1A 0E8

AND

JOSEPH P. HODYCH
Department of Earth Sciences, Memorial University of Newfoundland, St. John's, Nfld., Canada A1B 3X5

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Redbeds and minor volcanic rocks of the Early Silurian Wigwam Formation of the Botwood Group were sampled in the Exploits Subzone of the Dunnage Zone of central Newfoundland. At 18 sedimentary and 3 volcanic sites thermal demagnetization isolated a shallow northward (or southward) magnetization ($D = 356.0°$, $I = -16.8°$; $k = 12.9$; $\alpha_{95} = 9.2°$; $n =$ 21 site means) after correction for the tilt of bedding. A positive fold test demonstrates that this remanence predates regional Siluro–Devonian folding. A primary age for the remanence is indicated by positive conglomerate tests on volcanic clasts, and reversals correlated with sedimentary stratigraphy. Antiparallel remanence directions at sites of opposite polarity indicate that the primary remanence is not partially overprinted by a secondary component. The Exploits Subzone of the Dunnage Zone is interpreted to have been at low paleolatitudes (8.5°S ± 5°) during the Early Silurian. We reached a similar conclusion for the Notre Dame Subzone of the Dunnage Zone in a previous paleomagnetic study of Early Silurian redbeds and volcanic rocks of the King George IV Lake area. The low paleolatitudes obtained for the Wigwam and King George IV Lake rocks do not differ significantly and, hence, provide no evidence for a large ocean between the two subzones in the Early Silurian. However, the azimuthal orientation of the subzones at that time is still uncertain, and an ocean with its axis oriented in a north–south direction would not be detected by the paleomagnetic method.

Les couches rouges et quelques volcanites de la Formation de Wigwam, du Groupe de Botwood, ont été échantillonnés dans la Sous-zone d'Exploits de la Zone de Dunnage, dans la partie centrale de Terre-Neuve. Les échantillons de 18 sites sédimentaires et de 3 sites volcaniques ont été désaimantés par chauffage. Après correction pour l'inclinaison du litage ils ont fourni une aimantation nord (ou sud), isolée, peu profonde ($D = 356°$C; $I = -16,8°$; $k = 12,9$; $\alpha_{95} = 9,2°$; moyenne de $n = 21$ sites). Le test du pli positif démontre que la rémanence prédate l'événement régional de plissement siluro-dévonien. Un âge primaire pour la rémanence est indiqué par les tests positifs de conglomérat sur les fragments volcaniques, et les inversions sont corrélées avec la stratigraphie sédimentaire. Les directions antiparallèles de rémanence aux sites de polarité opposée révèlent que la rémanence primaire n'est aucunement surimprimée par une composante secondaire. La Sous-zone d'Exploits de la Zone de Dunnage est interprétée comme ayant été localisée à de basses paléolatitudes (8,5°S ± 5°) durant le Silurien précoce. Une étude paléomagnétique antérieure sur les couches rouges et les volcanites, d'âge silurien précoce, dans la région du lac King George IV a conduit à la même conclusion. Les basses paléolatitudes obtenues pour les roches de Wigwam et du lac King George IV ne diffèrent pas significativement et, par conséquent, elles ne fournissent aucune évidence de l'existence d'un large océan entre les deux sous-zones au Silurien précoce. Cependant, l'orientation azimutale des sous-zones à cette période est encore incertaine, et un océan avec son axe orienté dans une direction nord-sud ne pourrait pas être mis en évidence par la méthode paléomagnétique.

[Traduit par la rédaction]

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Introduction

There has long been agreement that the ophiolite suites of the Dunnage Zone of central Newfoundland (Fig. 1) represent vestiges of the floor of the Iapetus Ocean (the proto-Atlantic of Wilson 1966) and its marginal basins. The Humber Zone of western Newfoundland is thought to represent the North American continental margin of Iapetus, with most interpreting the Gander Zone of eastern Newfoundland as a sedimentary wedge at the opposite margin of Iapetus. There is good evidence that ophiolites of the Dunnage Zone were obducted onto the Humber Zone in the Mid-Ordovician (Williams and St-Julien 1982) and some evidence that ophiolites of the Dunnage Zone were obducted onto the Gander Zone in the Mid-Ordovician (Dec and Colman-Sadd 1990; Williams and Piasecki 1990). This implies that the Iapetus Ocean in Newfoundland was closing by the Mid-Ordovician, being consumed at both margins. However, there is disagreement as to when Iapetus closure was completed.

On one hand, McKerrow (1988) and Scotese and McKerrow (1990) suggested that the Iapetus in Newfoundland was still open in the Early Silurian and did not completely close until the Early Devonian Acadian orogeny. Williams *et al.* (1988) interpreted the Red Indian Line (Fig. 1) between the Notre Dame and Exploits subzones of the Dunnage Zone as a major terrane boundary. They proposed a geographical separation of the Notre Dame and Exploits subzones during the Early to Middle Ordovician to explain contrasting stratigraphy, lithology, faunas, plutonism, and other geological and geophysical characteristics across the Red Indian Line and its proposed extension southwestward to the Cape Ray fault (Fig. 1). For example, they pointed out that Early to Middle Ordovician brachiopod faunas found east of the Red Indian Line have oceanic island affinities in contrast with the North American affinities of a coeval Ordovician conodont fauna in the Notre Dame Subzone. Williams and O'Brien (1991) reported Middle Llandovery graptolites of European rather than North American affinity southeast of the Red Indian Line (at the locality marked by a star in Fig. 1), supporting the view that an ocean within the Dunnage Zone remained open in the Early Silurian.

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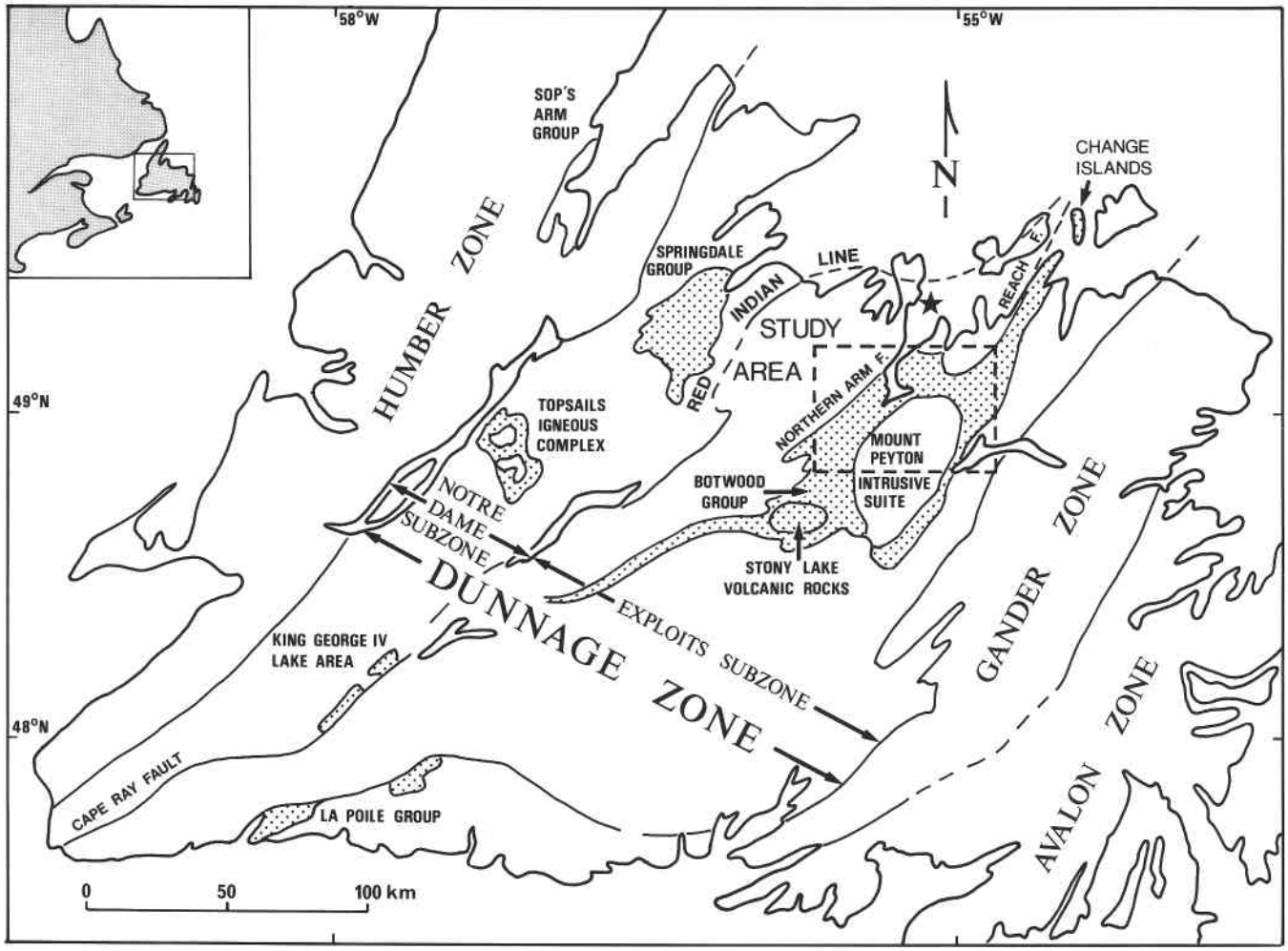


FIG. 1. Distribution of Early Silurian terrestrial volcanic and sedimentary rocks in Newfoundland. The study area is indicated and is shown in detail in Fig. 2. Tectonic–stratigraphic zone boundaries are those of Williams *et al.* (1988). ★, Llandovery graptolite locality of Williams and O'Brien (1991).

On the other hand, Early Silurian terrestrial sedimentary and volcanic rocks (Fig. 1) are widespread in the Dunnage and its neighbouring zones. This suggests that the Iapetus Ocean had almost completely closed in Newfoundland by the Early Silurian (Chandler *et al.* 1987). There is good evidence for the Early Silurian age of the Botwood Group (van der Pluijm *et al.* 1987), the Springdale Group (Chandler *et al.* 1987), the Topsails igneous complex (Whalen *et al.* 1987), the Sop's Arm Group (Dunning 1987), the Stony Lake volcanic rocks, the La Poile Group, and the redbeds and volcanic rocks (Bear Pond rhyolite) of the King George IV Lake area (Dunning *et al.* 1990). The completion of closure by the Silurian is also supported by recent U–Pb zircon dating, suggesting that much of the magmatism and deformation in central Newfoundland that had been ascribed to Devonian (Acadian) orogeny was due to an earlier Silurian orogeny (Dunning *et al.* 1990).

Paleomagnetism can be used to establish the paleolatitude and azimuthal orientation of the Notre Dame and Exploits subzones and, hence, to help constrain their relative location in the Early Silurian. For example, if the boundary between the subzones was oriented east–west in the Early Silurian, distinct paleolatitudes would demonstrate that the ocean existed between them, whereas similar paleolatitudes would indicate, within the uncertainty of the data, that no ocean was present. In this orientation, post-Early Silurian strike-slip motion along the boundary would not be detected. On the other hand, if the Early Silurian boundary was oriented north–south, post-Early Silurian strike slip would be detected, but the width of an ocean between the subzones could not be resolved.

Three previous paleomagnetic studies have been published for Early Silurian rocks of central Newfoundland, two from the Exploits Subzone and one from the Notre Dame Subzone. Unfortunately, the magnetizations obtained in the studies from the Exploits Subzone have not been shown to be of Early Silurian age, and the relative location of the two subzones at that time is unclear.

Early work of Lapointe (1979) showed that redbeds of the Wigwam Formation in the Exploits Subzone were magnetized before folding at a paleolatitude of 15°S ± 9° (uncertainty calculated herein). Gales *et al.* (1989) estimated a paleolatitude of 24°S ± 6° (uncertainty calculated herein) for the underlying Lawrenceton Formation volcanic rocks on the basis of a positive fold test between sites on Change Islands (Fig. 1) and sites of Lapointe (1979) at Grand Falls (Fig. 2). Neither study, however, demonstrated that the remanence of these rocks is primary. Indeed, the remanence on Change Islands was acquired after small-scale folding (although Gales *et al.*

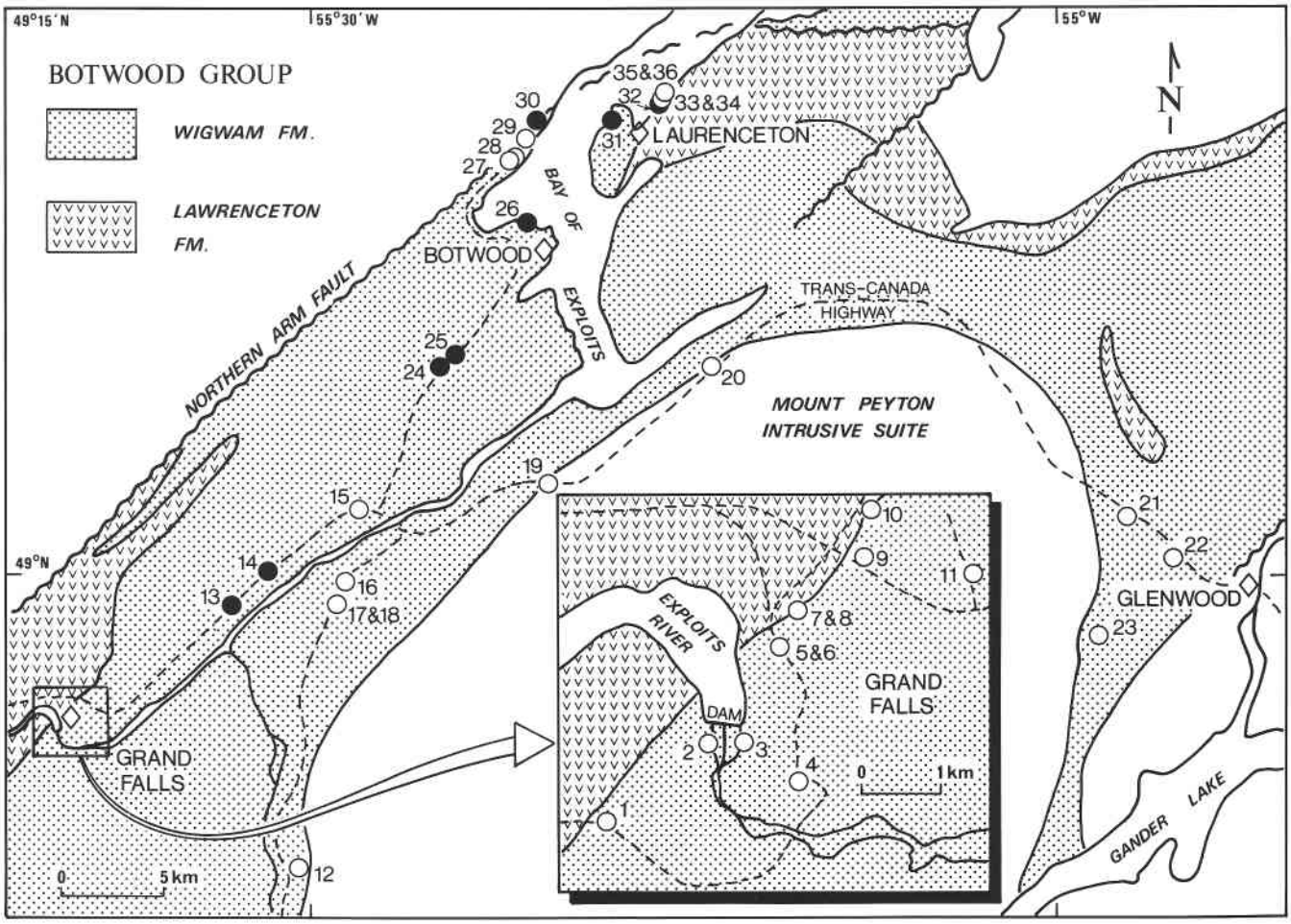


FIG. 2. Paleomagnetic sampling sites in the Wigwam Formation of central Newfoundland. Geology of the Botwood Group follows Andrews (1980) and Blackwood *et al.* (1984). The area near Grand Falls is enlarged in the insert. ○, stably magnetized sampling site; ●, unstably magnetized sampling site.

(1989) interpreted this folding to have occurred before the volcanic rocks cooled below their Curie temperature). Van der Pluijm *et al.* (1990) have accepted the 24°S paleolatitude for the Lawrenceton volcanic rocks and suggest that it is indistinguishable from the mid-Silurian paleolatitude for the adjacent North American craton margin. Unfortunately, cratonic paleopoles for this period are as yet rather poorly dated.

Using fold and conglomerate tests, Buchan and Hodych (1989) established that the 429^{+7}_{-3} Ma subaerial volcanic rocks (Bear Pond rhyolite of Dunning *et al.* 1990) and redbeds of the King George IV Lake area (Fig. 1) carry a primary remanence. This remanence indicates an Early Silurian paleolatitude of 0.5°N ± 6° for the Notre Dame Subzone of the Dunnage Zone. Our aim in the present paper is to provide an equally reliable Early Silurian paleolatitude for the Exploits Subzone of the Dunnage Zone by restudying the paleomagnetism of the Wigwam Formation.

Geological setting

Our study area (Fig. 1) lies in the middle of the Botwood Group, which extends over 200 km southwest from Change Islands. Near its type area around the Bay of Exploits, the Botwood Group consists of subaerial mafic and felsic volcanic rocks of the Lawrenceton Formation conformably overlain by the redbeds and minor volcanics of the Wigwam Formation. Our paleomagnetic sites are all in the Wigwam Formation (Fig. 2).

Near the south end of the Bay of Exploits, the Wigwam Formation contains limestone beds that bear late Llandovery to early Wenlock fossils (Williams 1972). Identification of a Wenlock to early Ludlow *Monograptus* near Glenwood is now suspect (Elliot *et al.* 1991). On Change Islands, the Lawrenceton Formation contains early and late Llandovery fossils (van der Pluijm *et al.* 1987), suggesting that the Wigwam Formation is not older than Llandovery. On the Port Albert Peninsula (southwest of Change Islands), the Wigwam Formation is intruded by dykes yielding a 422 ± 3 Ma U–Pb zircon age (Elliot *et al.* 1991). The Wigwam Formation appears to be overlain by Stony Lake volcanic rocks which yield a U–Pb zircon date of 423^{+3}_{-2} Ma (Dunning *et al.* 1990). Both of these U–Pb ages suggest that the Wigwam Formation is Wenlock or older, according to the time scale of Harland *et al.* (1990). Hence, the Wigwam Formation is very likely Wenlock or Llandovery, that is, Early Silurian. Its absolute age is likely not less than 422 ± 2 nor greater than 439 ± 2 Ma (the U–Pb zircon date for the early Llandovery of Tucker *et al.* 1990). Hence, 430 ± 10 Ma is a reasonable estimate of the Wigwam Formation age and its possible range.

Dunning *et al.* (1990) argue that much of the folding of the

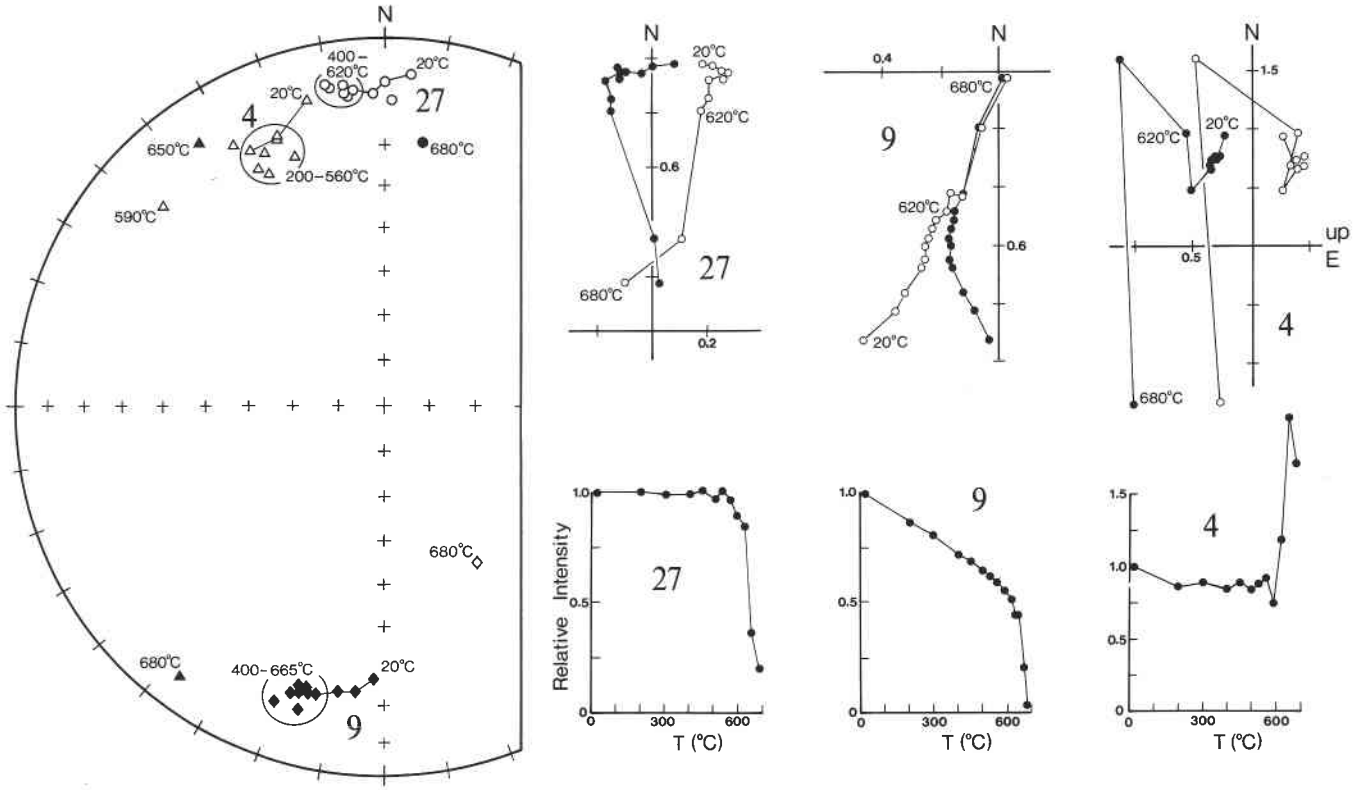


FIG. 3. Examples of thermal demagnetization of unmetamorphosed Wigwam Formation redbed specimens from sites 27, 9, and 4 after tilt correction. Directions are plotted on an equal-area net, with closed symbols indicating down directions and open symbols indicating up directions. On the orthogonal component plot, closed symbols represent projection onto the horizontal plane, whereas open symbols represent projection onto the north-up vertical plane. Data are plotted relative to initial intensity of natural remanent magnetization.

Botwood Group likely occurred in the Early Silurian soon after deposition, rather than in the Devonian, because the Botwood Group seems to be overlain by the relatively undeformed 423^{+3}_{-2} Ma Stony Lake volcanic rocks. This is supported by the observation that the Botwood Group folds appear to be cut by the Mount Peyton Intrusive Suite (Williams 1988, p. 62), whose oldest (gabbroic) phase yields an $^{40}Ar-^{39}Ar$ hornblende and biotite age of 420 ± 8 Ma (Reynolds *et al.* 1981).

Regional metamorphism of the Botwood Group is in the subgreenschist to low greenschist facies in the Grand Falls – Glenwood area (Kean and Mercer 1981; Blackwood 1982). Much of this metamorphism may be due to the Acadian (Devonian) orogeny. This is consistent with the 390 ± 15 Ma Rb–Sr age of the main (granitic) phase of the Mount Peyton Intrusive Suite (K. Bell and J. Blenkinsop, personal communication, reported by Reynolds *et al.* 1981). It is also consistent with the 385 ± 13 Ma total gas age yielded by the disturbed $^{40}Ar-^{39}Ar$ spectrum of the Lawrenceton Formation rhyolite at Grand Falls (Reynolds *et al.* 1981).

Sampling and laboratory procedure

Oriented block samples were collected from the Wigwam Formation at 36 sites (Fig. 2). Sampling was confined to localities where bedding attitude could be accurately measured.

Twenty-seven sites are in hematitic siltstones or fine-grained sandstones. At one site (9) sampling was conducted through a continuous 100 m stratigraphic section. Three additional sites (12, 19, and 20) were located in grey to black metasedimentary rocks within the metamorphic aureole of the Mount Peyton Intrusive Suite to allow comparison of the magnetization of metamorphosed and unmetamorphosed rocks. Volcanic flows were sampled at sites 6, 7, and 8, and attitudes were determined from adjacent sedimentary rocks. Conglomerate tests on volcanic clasts were attempted at sites 18, 33, and 35.

In the laboratory, cylindrical specimens (2.2 cm in length and 2.4 cm in diameter) were drilled from each sample. Alternating field (AF) demagnetization to 100 mT had little effect on the remanence of many sites. Hence, most specimens were thermally demagnetized in steps to 680°C. Usually, 10–12 steps were employed, using a commercial thermal demagnetizer with peak temperature control of ±20°C or a remodelled furnace with peak temperature control of ±5°C. Occasionally, up to 20 temperature steps were applied using the latter instrument. The remodelled furnace and the commercial spinner magnetometer with which specimen remanences were measured were housed in a magnetically shielded room and generally proved more efficient at isolating stable magnetizations.

Paleomagnetic results and analysis

Examples of thermal demagnetization of samples from Wigwam Formation redbeds are shown in Fig. 3. After removal of soft viscous remanences during cleaning to peak temperatures between 200 and 400°C, directions at most sites remained stable to temperatures between 600 and 680°C, and were directed either to the north or to the south (Fig. 3, sites 27 and 9). At several sites, severe chemical alterations during the heating experiments above about 600°C were indicated by

TABLE 1. Paleomagnetic results for Wigwam Formation redbeds and volcanic and metasedimentary rocks

Site	Lat. N	Long. W	St (°)	Di (°)	T	N (n)	K/E	D (°)	I (°)	D′ (°)	I′ (°)	k	α_{95} (°)
01	48°55.5′	55°40.9′	15	60	t	4(3)	2/1	206.3	14.2	187.9	16.6	36	21
02	48°55.8′	55°40.3′	320	50	t	5(3)	3/0	2.8	29.0	10.3	−7.6	7	49
03	48°55.8′	55°40.1′	280	60	t	5(3)	3/0	331.5	43.2	342.7	−8.8	39	20
04	48°55.7′	55°39.8′	352	56	t	5(5)	0/5	4.0	−8.0	352.0	−14.4	24	16
05	48°56.1′	55°39.8′	5	58	t	4(4)	4/0	24.1	−14.2	2.5	−23.6	15	24
06[a]	48°56.1′	55°39.8′	5	58	t	5(4)	4/0	26.0	−4.2	12.8	−20.0	31	17
07[a]	48°56.3′	55°39.7′	10	60	t	5(4)	4/0	22.5	−0.4	16.0	−11.1	61	12
08[a]	48°56.3′	55°39.7′	10	60	t	5(5)	5/0	16.0	−3.1	10.3	−6.8	28	15
09N	48°56.4′	55°39.3′	31	75	t	5(5)	5/0	58.2	−31.1	3.5	−30.8	13	22
09S	48°56.4′	55°39.3′	31	75	t	10(10)	10/0	225.0	30.2	183.8	19.4	14	13
10	48°56.6′	55°39.6′	30	80	t	5(4)	0/4	212.1	24.8	185.8	6.1	85	10
11	48°56.3′	55°38.7′	22	70	t	5(3)	0/3	206.5	40.7	163.9	16.2	79	14
12[b]	48°52.1′	55°30.9′	25	42	f	3(3)	3/0	267.7	86.3	117.7	51.2	16	32
15	49°01.7′	55°27.8′	215	75	t	6(3)	0/3	28.4	29.8	4.6	1.9	21	27
16	48°59.8′	55°28.7′	200	75	t	6(6)	3/3	157.8	10.8	200.8	43.4	21	15
17	48°59.2′	55°29.1′	220	55	t	5(5)	0/5	183.8	16.2	212.8	38.7	36	13
19[b]	49°02.2′	55°19.3′	15	45	f	5(4)	4/0	42.7	66.2	80.4	31.0	19	22
20[b]	49°05.4′	55°14.0′	65	10	f	5(5)	5/0	327.5	76.4	308.5	86.1	27	15
21	49°01.7′	54°57.3′	220	90	t	5(4)	0/4	346.1	42.3	342.9	−36.7	8	36
22	49°00.6′	54°55.2′	180	110	t	5(3)	3/0	348.1	19.1	345.4	−17.1	58	16
23	48°58.5′	54°58.3′	200	80	t	4(4)	4/0	314.7	34.7	316.4	−39.5	15	25
27	49°10.7′	55°22.1′	20	60	t	6(4)	4/0	10.0	−35.3	344.6	−9.6	23	20
28	49°10.8′	55°22.0′	40	73	t	5(5)	5/0	117.4	87.6	129.4	14.6	15	26
29	49°11.3′	55°21.3′	30	15	t	5(5)	0/5	334.7	−13.9	333.6	−1.5	11	25
34	49°12.6′	55°15.8′	215	62	t	6(5)	2/3	5.5	36.7	349.4	−3.9	8	29
36	49°12.7′	55°15.7′	215	60	t	5(5)	5/0	19.8	32.2	0.0	4.2	18	19
Mean of redbed and volcanic sites[c]						21		11.2	−6.7			4.5	17.0
Mean of redbed and volcanic sites[d]						21				356.0	−16.8	12.9	9.2

NOTES: Only stably magnetized sites are listed. Two sites (2 and 21) with large uncertainties ($\alpha_{95} \geq 30°$) are not included in the mean statistics. Metasedimentary sites (12, 19, and 20) with steep down overprint directions are not included in mean statistics. Lat. and Long., latitude and longitude, respectively, of the sampling locality; St and Di, strike and dip, respectively, of bedding; T, treatment used: either thermal (t) or alternating field (f); *N*, number of samples measured; *n*, number of samples used in statistics (unstable samples were excluded); *K/E*, number of sample direction determined by Kirschvink/end-point methods; *D* and *I*, mean declination and inclination, respectively, of remanence before tilt correction; *D′* and *I′*, mean declination and inclination, respectively, after tilt correction; *k*, precision parameter; α_{95}, radius of the circle of 95% confidence about the mean direction.

[a]Volcanic sites.

[b]Metasedimentary sites.

[c]Corresponding paleopole before tilt correction: 36.7°N, 110.5°E; dm = 17.1°; dp = 8.6°. dm and dp are the semiaxes of the oval of 95% confidence about the mean paleopole.

[d]Corresponding paleopole after tilt correction: 32.3°N, 129.2°E; dm = 9.5°; dp = 4.9°.

large increases in intensity and obscured the demagnetization process (Fig. 3, site 4). Blocking temperatures suggested that hematite is the sole carrier of stable remanence in most redbed samples. AF demagnetization to 100 mT had little effect on the magnetization of these samples. At eight redbed sampling sites (13, 14, 24, 25, 26, 30, 31, and 32) magnetization directions did not remain stable on thermal demagnetization. At two sites (2 and 21) where individual sample directions were stable, directions among samples were too scattered ($\alpha_{95} \geq 30°$) to be useful in calculating a Wigwam Formation paleopole.

At site 9, sixteen samples were measured from a continuous 100 m stratigraphic section of Wigwam redbeds deposited on volcanic flows of the Lawrenceton Formation. A sequence of 3 reversals was observed through the section. At the bottom, south-directed remanence directions were obtained from hematitic siltstone, which penetrates fractures within the upper 3 m of Lawrenceton volcanic flows, and from about 30 m of overlying bedded hematitic sandstone. Above are sandstone units of about 35, 20, and 10 m thickness, which carry north-, south-, and north-directed magnetizations, respectively. For purposes of statistical analysis north- and south-directed remanences at this locality were averaged separately and reported as sites 9N and 9S (Table 1).

Samples from sites 12, 19, and 20 in grey metasedimentary rocks within the metamorphic aureole of the Mount Peyton Intrusive Suite (Fig. 2) exhibited steep downward remanence directions and low coercivities during AF demagnetization (Fig. 4). As discussed below, these sites likely carry secondary magnetizations acquired either in the recent Earth's magnetic field or at the time of emplacement of the Mount Peyton Intrusive Suite.

Sites 6, 7, and 8 are located in volcanic flows of the Wigwam Formation which occur stratigraphically less than 200 m above the Lawrenceton–Wigwam contact in Grand Falls (see Fig. 2). The volcanic flows rest on bedded sedimentary rocks, from which their attitudes were obtained. The flow at site 8 is located above the flow of site 7, with a thin sedimentary layer between them. The magnetization of the volcanic rocks was well defined (Fig. 5), and blocking temperatures indicated that the stable component is carried by magnetite in some samples and by both magnetite and hematite in others. After tilt correction the direction of magnetization of

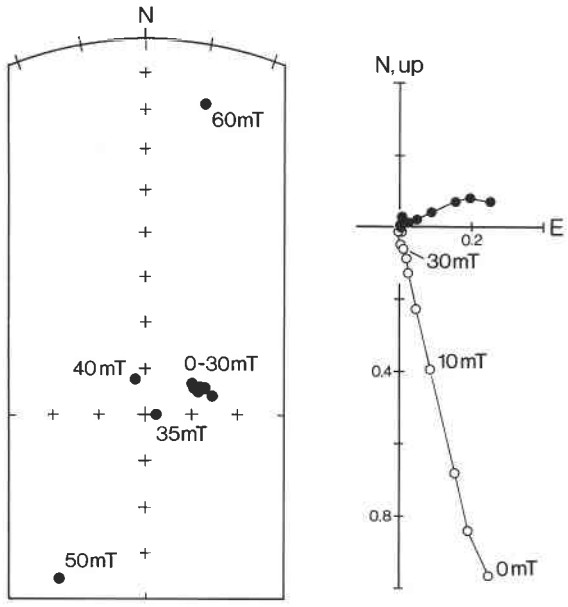


FIG. 4. Example of AF demagnetization of a Wigwam Formation redbed sample from the metamorphic aureole of the Mount Peyton Intrusive Complex before tilt correction. Symbols are as in Fig. 3.

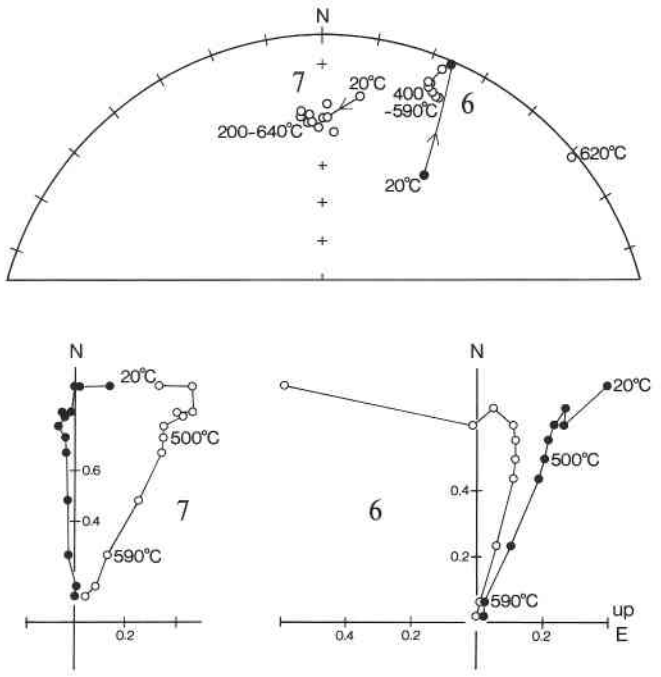


FIG. 5. Examples of thermal demagnetization of Wigwam Formation volcanic specimens at sites 7 and 6 after tilt correction. Symbols are as in Fig. 3.

the volcanic rocks agrees with tilt-corrected remanence directions of Wigwam Formation sedimentary rocks. In particular, the porphyritic volcanic rocks at site 6 and hematitic siltstones of site 5, immediately below, carry a magnetization of similar direction (Table 1).

Mean paleomagnetic directions for stably magnetized redbed and volcanic sites are summarized in Table 1. For most sites, magnetization directions for individual specimens were established using the least-squares fit method of Kirschvink (1980). At several sites, however, the Kirschvink method was difficult to employ because heating above 600°C caused severe chemical alterations that obscured the stable component, often before its intensity has substantially decreased. In such cases, a stable end point was calculated as the mean of stable directions observed after elimination of soft, viscous components, and before significant chemical alteration.

Timing of remanence acquisition

Fold test

We have tested whether the remanence of the Wigwam Formation was acquired before or after Siluro-Devonian folding. For statistical purposes, data from 18 redbed and 3 volcanic sites were used. Excluded were the 2 sites (2 and 21) with large uncertainties in direction of magnetization ($\alpha_{95} \geq 30°$), 3 metasedimentary sites (12, 19, and 20) with secondary overprints, and the 8 unstably magnetized sites. Before tilt correction there was considerable scatter of the 21 mean site directions (Table 1; Fig. 6*a*). Reversing the south-directed magnetizations yielded an overall in situ mean direction of $D = 11.2°$, $I = -6.7°$; $k = 4.5$; $\alpha_{95} = 17.0°$.

After tilt correction (assuming horizontal fold axes) the scatter was significantly reduced (Fig. 6*b*), and the mean direction of $D = 356.0°$, $I = -16.8°$; $k = 12.9$; $\alpha_{95} = 9.2°$ was better constrained. Using the criteria of McElhinny (1964) the data passed the fold test (Graham 1949) at the 99% confidence level (precision parameter $k = 4.5$ *in situ* and 12.9 after tilt correction), demonstrating that the remanence predates folding. The *f*-distribution test of McElhinny has been criticized as too stringent, and an alternative test has been proposed by McFadden (1990). Applying the McFadden correlation test also demonstrated that the remanence in the Wigwam Forma-

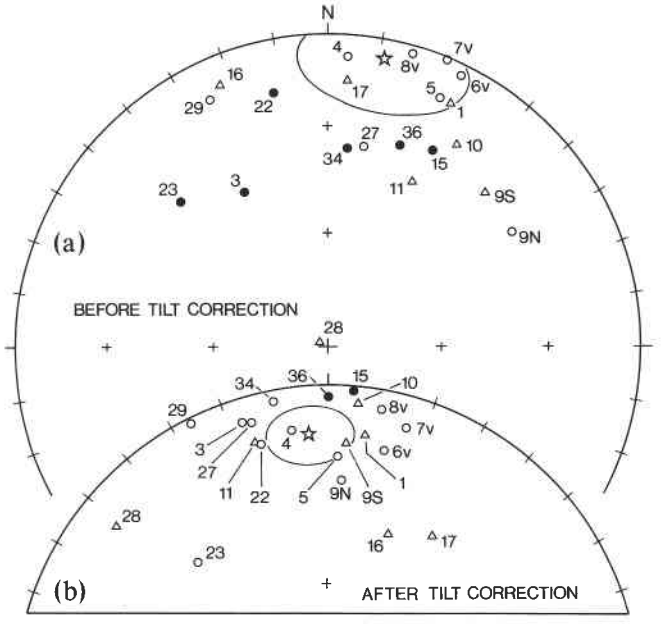


FIG. 6. Fold test for stably magnetized Wigwam Formation redbed and volcanic sites. Site directions and their mean (☆) with its circle of 95% confidence are shown (*a*) before tilt correction and (*b*) after tilt correction. ○, ●, north-seeking magnetizations; △, ▲, south-seeking magnetizations. Sites in volcanic rocks are labelled V. Open symbols, directions plotted on the upper hemisphere; closed symbols, directions plotted on the lower hemisphere.

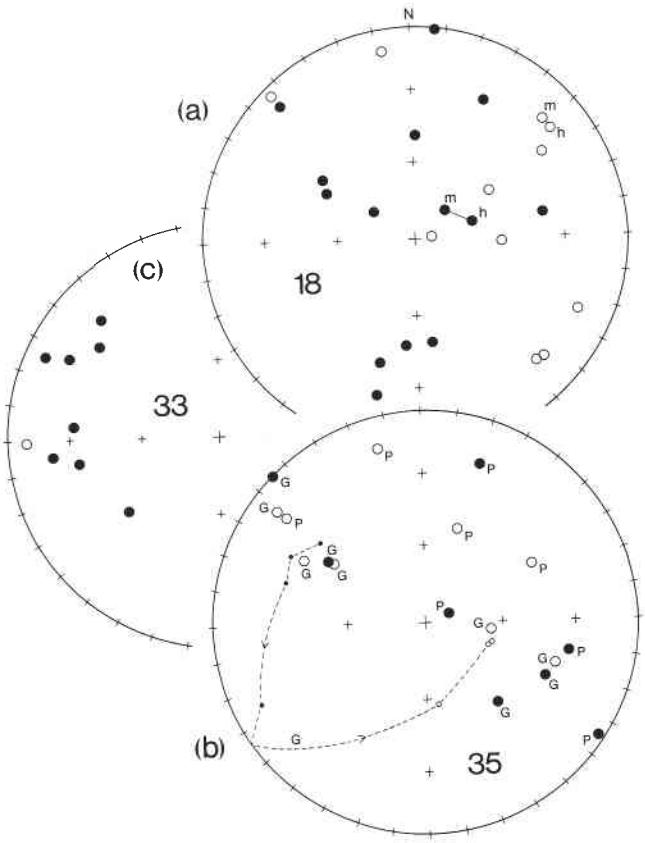


FIG. 7. Conglomerate tests based on thermal demagnetization data at (*a*) sampling site 18, (*b*) site 35, and (*c*) site 33. Magnetization directions determined using Kirschvink analysis are shown for individual volcanic clasts after tilt correction of the host strata. The broken line in Fig. 7*b* illustrates the systematic swing in magnetization direction of one sample during stepwise demagnetization between 0 and 560°C. In Fig. 7*a* rhyolite clasts carry identical magnetite and hematite remanence directions except for two samples where magnetite (m) and hematite (h) directions are shown separately. In Fig. 7*b* both grey (G) and purple (P) clasts carry stable hematite remanences. Symbols are as in Fig. 3.

tion predates Siluro-Devonian folding (test statistic $\xi_1 = 12.6$ *in situ* and 3.5 after tilt correction, with the minimum ξ_1 occurring at 100% untilting).

Conglomerate tests

Conglomerate tests were attempted at site 18, about 12 km northeast of Grand Falls, and at sites 33 and 35, along the coast just to the northeast of Laurenceton (Fig. 2).

At site 18, 23 volcanic clasts were collected from an intraformational conglomeratic bed overlying the hematitic siltstones of site 17, where a consistent, stable direction of magnetization was obtained (Table 1). Individual conglomerate clasts carried stable directions of magnetization, but directions between clasts were highly scattered (Fig. 7*a*). In most clasts both magnetite and hematite components were observed (e.g., Fig. 8) and were indistinguishable in direction. However, a small but measurable difference was seen in two of the clasts (see Fig. 7*a*).

The directions of magnetization from this conglomerate passed Watson's (1956) randomness test ($R_0/R = 7.7/5.3$) at the 95% confidence level (Irving 1964, Table 4.4). Thus the conglomerate test was positive, demonstrating that the rhyolite clasts carry a predepositional remanence.

Additional support for this interpretation was obtained from a positive conglomerate test at site 35 (Fig. 7*b*). Some of the 18 volcanic clasts are grey, and others purple, with the latter likely derived from oxidized flow tops. The remanence in these clasts is usually carried by hematite and is generally quite stable. An exception was one clast that exhibited a dramatic swing in remanence direction between 20 and 560°C (Fig. 7*b*). The directions from this conglomerate also passed the Watson randomness test at the 95% confidence level ($R_0/R = 6.6/4.3$), indicating that the remanence of the conglomerate clasts is predepositional. Hematitic siltstones (site 36) that overlie the conglomerate at site 35 gave a consistent, stable remanence direction.

The two positive conglomerate tests for volcanic clasts at widely separated sites 18 and 35 indicate that Wigwam Formation volcanic rocks (e.g., Fig. 5) retain a primary remanence. In addition, the agreement between directions of magnetization observed at volcanic and sedimentary sites (see Fig. 6*b*) suggests that the remanence of the redbeds is also primary.

The nature of the Wigwam Formation data rules out the possibility that a significant partial overprint has affected the primary remanence in these rocks. Starkey and Palmer (1970) have pointed out that a small partial overprint superimposed along a consistent direction upon a random scatter of predepositional remanence directions in conglomerate clasts is not readily detected by the randomness test. However, a partial overprint should result in systematic directional swings as individual conglomerate clasts are demagnetized since blocking temperature spectra for the two magnetizations are unlikely to be identical (especially in clasts containing both magnetite and hematite, as at site 18). Also, a partial overprint should cause asymmetry in the reversal pattern observed between stably magnetized sites. A partial overprint in rocks of the Wigwam Formation seems unlikely because almost all specimens at sites 18 and 35 carry stable directions of magnetization, and mean directions for sites magnetized to the north and south are antiparallel within the limits of statistical uncertainty.

At conglomerate site 33, located at the base of the Wigwam Formation or the top of the Lawrenceton Formation near Laurenceton, directions of magnetization from nine purple volcanic clasts were fairly stable. Although the number of samples is insufficient for a proper conglomerate test, the clasts all had a shallow westward direction of magnetization after tilt correction (Fig. 7*c*). Further inspection of this sampling locality indicated that clasts are not rounded and that the sedimentary host may simply infill a fractured flow top. Hence, these samples, although somewhat displaced from their original positions, are unlikely to represent random sample orientations necessary for a conglomerate test. Nevertheless, the shallow westward mean direction of magnetization is difficult to explain. It does not correspond to the shallow north- or south-directed remanence observed at stable Wigwam Formation sites; nor does it correspond to remanence directions reported for the Lawrenceton Formation by Lapointe (1979) and Gales *et al.* (1989). Indeed, hematitic siltstones at site 34, immediately above the "conglomerate" at site 33, yielded a typical north-seeking Wigwam Formation magnetization (Table 1).

Discussion

A positive fold test for Wigwam Formation sedimentary rocks and volcanic flows demonstrates that, with the exception of sites in metasedimentary rocks adjacent to the Mount

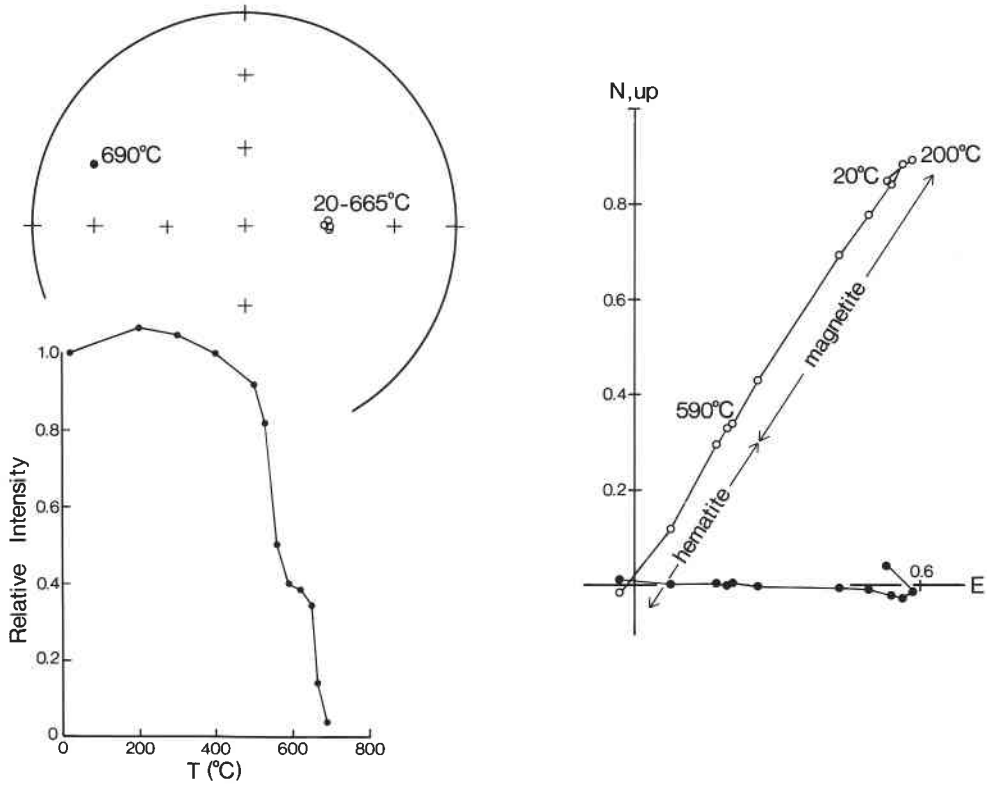


FIG. 8. Example of thermal demagnetization of a clast from conglomerate site 18 of Fig. 7*a* illustrating the presence of both magnetite and hematite components. Symbols are as in Fig. 3.

Peyton Intrusive Suite, the remanence of stably magnetized sites was aquired before folding. This confirms the earlier work of Lapointe (1979), based on a more restricted data set. Folding is thought to largely predate emplacement of the 423^{+3}_{-2} Ma Stony Lake volcanic rocks and the 420 ± 8 Ma mafic phase of the Mount Peyton Intrusive Suite (Dunning *et al.* 1990). If so, the remanence of the Wigwam Formation must date from the Early Silurian.

Positive conglomerate tests at two widely separated localities establish a predeposition age for the remanence of the volcanic clasts in the conglomerate. Hence, the remanence of the three Wigwam Formation volcanic sites sampled in this study is interpreted as primary. Furthermore, the agreement between the magnetization direction of volcanic and sedimentary rocks after unfolding indicates that the magnetization of the Wigwam Formation redbeds also dates from the time of deposition in the Early Silurian.

Additional support for a primary origin for the magnetization of Wigwam Formation sedimentary rocks comes from a series of reversals obtained through a steeply dipping continuous stratigraphic sequence (site 9) at Grand Falls. After tilt correction, the directions in these hematitic siltstones and sandstones are in good agreement with other tilt-corrected results from Wigwam Formation sedimentary rocks.

Finally, there is no evidence of a partial remagnetization of the Wigwam Formation sedimentary and volcanic rocks. Such an overprint would produce asymmetry between magnetization directions of opposite polarity. However, normal and reversed Wigwam Formation mean directions are antiparallel within the statistical uncertainty of the data.

Sedimentary rocks at three Wigwam Formation sites within the metamorphic aureole of the Mount Peyton Intrusive Suite have been altered to a grey or black colour. Magnetite, the principal remanence carrier, is magnetized in a steep downward direction before untilting, similar to directions reported by Lapointe (1979) in granitic phases of the Mount Peyton Intrusive Suite. The metasedimentary rocks could have remagnetized at the time that the granite acquired its remanence. Alternatively, the remanences of the Wigwam Formation metasedimentary rocks and the granite of the Mount Peyton Intrusive Suite, which are both of rather low coercivity, may have remagnetized in the recent Earth's field.

The Mount Peyton Intrusive Suite appears to cut across fold axes of the Wigwam Formation and is thus interpreted to have been intruded after folding (Williams 1988). Hence, the positive fold test also implies that heating associated with intrusion of the Mount Peyton Intrusive Suite did not result in remagnetization of the 21 unmetamorphosed paleomagnetic sites used in the test. The positive conglomerate tests further support this conclusion.

The primary paleomagnetic direction determined for the Wigwam Formation (D = 356.0°, I = −16.8°; k = 12.9; α_{95} = 9.2°) corresponds to a paleopole of 129.2°E, 32.3°N; dm = 9.5°; dp = 4.9° and a paleolatitude of 8.5°S ± 5°. Thus, the northern portion of the Exploits Subzone is interpreted to have been at low paleolatitudes during the Early Silurian. A similar conclusion was reached for the southwestern portion of the Notre Dame Subzone in an earlier study of Early Silurian redbeds and volcanic rocks of the King George IV Lake area (Buchan and Hodych 1989). There a paleolatitude of 0.5°N ± 6° was obtained, and a primary remanence was confirmed from a positive conglomerate test.

These two paleolatitude determinations do not differ significantly. Thus, the paleomagnetic results offer no support for a large open ocean between the Notre Dame and Exploits subzones in the Early Silurian. As described earlier, however, an

ocean that is not detectable in the paleomagnetic results is possible if the boundary between the subzones was oriented approximately north–south.

Unfortunately, the Early Silurian orientation of the subzones is still unclear. The paleomagnetic declination for the King George IV Lake area is about 30 degrees east of that of the Wigwam Formation, suggesting that relative rotation has occurred between these areas since Early Silurian deposition. Much of this difference could be due to dextral strike slip along the faults that appear to bound the King George IV Lake area to the northwest and southeast (Fig. 2 of Buchan and Hodych 1989).

In addition, the paleomagnetic data do not preclude an Early Silurian ocean elsewhere in Newfoundland. For example, Kusky *et al.* (1987) suggest that the Wigwam Formation of the Botwood Group and the Early Silurian redbeds of the King George IV Lake area were deposited in pull-apart basins which opened by dextral strike slip along a fault system extending from the Reach and Northern Arm faults of north-central Newfoundland to the Cape Ray fault of southwestern Newfoundland. According to their model these basins formed as a result of oblique subduction of an ocean located to the east along the boundary between the Dunnage and Gander zones.

A low paleolatitude for the Dunnage Zone during the Early Silurian, as suggested by the paleomagnetism of the Wigwam Formation and the redbeds and volcanic rocks of the King George IV Lake area, conflicts with the interpretation of van der Pluijm *et al.* (1990). They propose a significantly higher paleolatitude for the Dunnage Zone on the basis of paleomagnetic data of Lapointe (1979) and Gales *et al.* (1989) from the Lawrenceton Formation of the Botwood Group. They further indicate that the paleolatitude for the Lawrenceton Formation (24°S) agrees with the mean paleolatitude for the stable North American craton (~25°S) as calculated from the summary of North American poles in Van der Voo (1988). The reliability of the Lawrenceton Formation data and reference results from the North American craton will now be discussed.

The Lawrenceton Formation volcanic rocks lie conformably beneath the Wigwam Formation. Gales *et al.* (1989) describe a negative fold test between two sampling sites on either limb of a mesoscopic fold (1.5 m in size) on Change Islands. Nevertheless, they interpret the remanence to be "nearly primary" and suggest that the small fold "was formed during deposition and while the rocks were still above their Curie point" (Gales *et al.* 1989, p. 303). Alternatively, these data could be interpreted to imply that the remanence is secondary and was acquired after regional folding.

For a regional fold test, Gales *et al.* (1989) compare their results from the Change Islands area with those of Lapointe (1979) from three sites of Lawrenceton Formation volcanics at a single locality in Grand Falls. They interpret the combined data as a positive fold test, implying that the remanence of the Lawrenceton Formation volcanic rocks in both areas was acquired prior to folding at a paleolatitude of about 24°. However, the three sites of Lapointe (1979) were collected from a single geographic locality and could even represent a single flow. In any case, they are unlikely to average out secular variation, casting doubt on the validity of the fold test. In our opinion, this uncertainty and the uncertainty noted above concerning the primary nature of the Lawrenceton remanence from Change Islands suggest that, until further studies are available, the results from the Lawrenceton Formation volcanic rocks should be used with extreme caution.

An updated summary of paleopoles from the North American craton has recently been published (Van der Voo 1990). Fourteen paleopoles are listed for the Late Ordovician to Late Silurian – Early Devonian period. However, positive fold tests have been obtained for only six of these. Five poles from three units have been shown to predate Permian folding. They include those for the north and south limbs of the Late Silurian Bloomsburg Formation, the mid-Silurian Rose Hill Formation, and the north and south limbs of the Late Ordovician Juniata Formation. Only one pole (that for the mid-Silurian or Late Silurian Wabash Reef Limestone) is known to predate Devonian or earlier deformation. None of the 14 poles has been demonstrated primary on the basis of a conglomerate test. Until better age control on the remanence of cratonic rocks is available, it is not possible to determine a reliable Early Silurian paleolatitude for cratonic North America.

In conclusion, data from the Wigwam Formation (this study) and the King George IV Lake area (Buchan and Hodych 1989) yield the only reliably dated paleopoles for the Dunnage Zone of Newfoundland during the Early Silurian. Early Silurian paleolatitudes for both the Exploits and Notre Dame subzones seem to be more equatorial than those for the adjacent North American craton, but this is uncertain since the cratonic paleolatitudes are inferred from remanences that are not proven to be primary.

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